

The Topology of Fiber Bundles

Labix

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1 Introduction to Principal Bundles

1.1 Fiber Bundles with a Group Action

We would now like to enrich the structure of a fiber bundle with a group action on the fibers. Recall that a topological group is a group with the structure of a topology such that multiplication $(g_1, g_2) \mapsto g_1 g_2$ and the inverse $g \mapsto g^{-1}$ are both continuous maps.

Definition 1.1.1 (G-Atlas)

Let $\pi : E \rightarrow B$ be a fiber bundle with fiber F . Let G be topological group. A G -atlas on the fiber bundle consisting of the following data:

- A continuous and faithful action of G on F (so that G injects into $\text{Aut}(F)$).
- A set of local trivialization charts

$$\{(U_k \subseteq B, \varphi_k : \pi^{-1}(U_k) \rightarrow U_k \times F) \mid k \in I\}$$

(where $B = \bigcup_{k \in I} U_k$ and φ_k is a homeomorphism for all $k \in I$).

- A function of sets $g_{ij} : U_i \cap U_j \rightarrow \text{Aut}(F)$ for $i, j \in I$ called transition functions such that the following conditions are satisfied:
 - For $i, j \in I$, the map g_{ij} lies in the injective image of G in $\text{Aut}(F)$.
 - For $i, j \in I$, the map $g_{ij} : U_i \cap U_j \rightarrow G$ is continuous.
 - The transition maps $\psi_{ji} = \varphi_j \circ \varphi_i^{-1} : (U_i \cap U_j) \times F \rightarrow (U_i \cap U_j) \times F$ is given by

$$\psi_{ji}(b, f) = \varphi_j \circ \varphi_i^{-1}(b, f) = (b, g_{ij} \cdot f)$$

Notice that for non empty intersections $U_i \cap U_j$ for U_i, U_j open sets in B , there is a well defined homeomorphism

$$\varphi_j \circ \varphi_i^{-1} : (U_i \cap U_j) \times F \rightarrow (U_i \cap U_j) \times F$$

This is reminiscent of properties of an atlas on M .

Definition 1.1.2 (Equivalent G -Atlas) Let G be a topological group. Let (F, E, B) be a fiber bundle and let $\{(U_i, \varphi_i \mid i \in I\}$ and $\{(V_j, \phi_j) \mid j \in J\}$ be two G -atlas on the fiber bundle. We say that they are equivalent if for all $x \in U_i \cap V_j$,

$$\bar{g}_{ji}(x) = \phi_{j,x} \circ \phi_{i,x}^{-1}$$

is an element of G and the map

$$\bar{g}_{ji} : U_i \cap V_j \rightarrow G$$

defined by $x \mapsto \bar{g}_{ji}(x)$ is continuous.

It is clear that this is an equivalent relation. It is reflexive directly from definitions of a G -atlas. ???

Definition 1.1.3 (G-Bundle) Let G be a topological group. A G -bundle is a fiber bundle (F, E, B) together with an equivalence class of G -atlas. In this case, G is said to be the structure group of the fiber bundle.

Similar to the case of manifolds, an maximal atlas is exactly such an equivalence class of atlases. Therefore one can also think of a G -bundle as a fiber bundle equipped with a maximal G -atlas.

Example 1.1.4

Let (F, E, B) be a vector bundle. Then (F, E, B) is a $GL(F)$ -bundle.

A morphism of G -bundles must now take into account of the group G .

Definition 1.1.5 (Morphisms of G -Bundles) Let G be a topological group. Let (F, E_1, B_1) and (F, E_2, B_2) be two G -bundles with common fiber. A morphism of G -bundles is a morphism of

bundles

$$(\bar{h} : E_1 \rightarrow E_2, h : B_1 \rightarrow B_2)$$

such that the following are true.

Let $\{(U_i, \varphi_i) \mid i \in I\}$ and $\{(V_j, \phi_j) \mid j \in J\}$ be local trivialities.

- For all $x \in U_i \cap h^{-1}(V_j)$, the map

$$\overline{g_{ji}}(x) = \phi_{j,x} \circ h|_{F_x} \circ \varphi_{i,x}^{-1} : F \rightarrow F$$

is an element of G .

- The map

$$\overline{g_{ij}} : U_i \cap h^{-1}(V_j) \rightarrow G$$

defined by $x \mapsto \overline{g_{ji}}(x)$ is continuous.

Since morphisms of fiber bundles preserve fibers, morphisms of G -bundles also preserve fibers.

It is easy to see that the mapping transformations $\widetilde{g_{ij}}$ satisfy the following two relations:

- $\widetilde{g_{jk}}(x) \cdot \widetilde{g_{ij}}(x) = \widetilde{g_{ik}}(x)$ for all $x \in U_i \cap U_j \cap h^{-1}(V_k)$
- $g'_{jk}(h(x)) \cdot \widetilde{g_{ij}}(x) = \widetilde{g_{ik}}(x)$ for all $x \in U_i \cap h^{-1}(V_j \cap V_k)$

g'_{jk} here refers to the transition charts in (F, E_2, B_2) .

1.2 The Cocycle Condition

Definition 1.2.1 (The Cocycle Condition)

Let X be a space. Let G be a topological group. Let $\{U_i \mid i \in I\}$ be an open cover of X . Let $\{g_{ij} : U_i \cap U_j \rightarrow G \mid i, j \in I\}$ be a collection of maps. We say that they satisfy the cocycle condition if

$$g_{ki}(x) = g_{kj}(x)g_{ji}(x)$$

for all $x \in U_i \cap U_j \cap U_k$.

Lemma 1.2.2

Let X be a space. Let G be a topological group. Let $\{U_i \mid i \in I\}$ be an open cover of X . Let $\{g_{ij} : U_i \cap U_j \rightarrow G \mid i, j \in I\}$ be a collection of maps that satisfy the cocycle condition. Then the following are true.

- $g_{ji}(x) = (g_{ij}(x))^{-1}$ for all $x \in U_i \cap U_j$.
- $g_{ii}(x) = 1$ for all $x \in U_i$.

Proposition 1.2.3

Let X, F be spaces. Let G be a topological group. Let $p : E \rightarrow X$ be a G -bundle with fiber F . Let g_{ij} be the transition functions of the bundle. Then $\{g_{ij} \mid i, j \in I\}$ satisfy the cocycle condition.

1.3 Principal G-Bundles

Let us recall some definitions from Groups and Rings. Let G be a group acting on a set X . We say that a group action is free if $g \cdot x = x$ for all $x \in X$ implies that $g = 1_G$. A group action is transitive if for any $x, y \in X$ there exists $g \in G$ such that $g \cdot x = y$.

Definition 1.3.1 (Principal Bundles)

Let G be a topological group. A principal G -bundle is a G -bundle (F, E, B) together with a continuous group action G on E such that the following are true.

- The action of G preserves fibers. This means that $g \cdot x \in E_b$ if $x \in E_b$. (This also means that G is a group action on each fiber)
- The action of G on each fiber is free and transitive

- The map $G \rightarrow F$ defined by sending $g \mapsto g \cdot x$ for any $x \in X$ is a homeomorphism.
- Local triviality condition: Each local triviality map $\varphi_U : p^{-1}(U) \rightarrow U \times F$ are G -equivariant maps.

Recall that even if G acts freely and transitively on X , G is still not homeomorphic to X . This is because the map $G \rightarrow X$ defined by $g \mapsto g \cdot x$ for any $x \in X$ is continuous and bijective. One usually requires that either G is compact or more in general, the above map is an open map then G will be homeomorphic to X .

Example 1.3.2

The following are true regarding the stiefel and grassmannian manifolds.

- The projection map $V_k(\mathbb{R}^n) \rightarrow \text{Gr}_k(\mathbb{R}^n)$ is a principal $\text{GL}(k, \mathbb{R})$ -bundle.
- The projection map $V_k(\mathbb{C}^n) \rightarrow \text{Gr}_k(\mathbb{C}^n)$ is a principal $\text{GL}(k, \mathbb{C})$ -bundle.

Example 1.3.3

The projection map $S^{2n+1} \rightarrow \mathbb{CP}^n$ is a principal S^1 -bundle.

Lemma 1.3.4

Let G be a topological group. Let $p : E \rightarrow B$ be a principal G -bundle. Then p is a G -bundle.

For those who know what homogenous spaces are, principal bundles are G -bundles such that F is a principal homogenous space for the left action of G itself.

Proposition 1.3.5 Let G be a topological group. Let $p : E \rightarrow B$ be a principal G -bundle. Then there is a homeomorphism

$$E/G \cong B$$

where E/G is the orbit space.

Conversely, given a continuous group action on a space, we can ask in what conditions will the space be a principal bundle over the orbit space.

Proposition 1.3.6 Let X be a space with a free G action. Let $p : X \rightarrow X/G$ be the projection map to the orbit space. If for all $x \in X/G$, there is a neighbourhood U of x and a continuous map $s : U \rightarrow X$ such that $p \circ s = \text{id}_U$, then $(G, X, X/G)$ is a principal G -bundle.

This proposition essentially means that if for each point in X/G , there is a local section, then it is sufficient for X to be a principal G bundle over X/G .

Proposition 1.3.7 Let X be a space. Let $p : \tilde{X} \rightarrow X$ be a covering space. If p is a normal covering space, then p is a principal G -bundle for some group G .

Definition 1.3.8 (Morphism of Principal Bundles) Let G be a topological group. Let $p_1 : E_1 \rightarrow B_1$ and $p_2 : E_2 \rightarrow B_2$ be two principal G -bundle. A morphism of principal G -bundles is a morphism of fiber bundles $(\tilde{f}, f)$ such that $\tilde{f}$ is a G -equivariant map.

Definition 1.3.9 (Isomorphism of Principal Bundles) Let G be a topological group. Let $p_1 : E_1 \rightarrow B_1$ and $p_2 : E_2 \rightarrow B_2$ be two principal G -bundle. We say that they are isomorphic if there exists a morphism of principal bundles $(\tilde{f}, f)$ such that it is an isomorphism in the sense of 1.1.6.

Principal bundles enjoy a very unique property. If one can construct a morphism between two principal bundles then the structure of a fiber as the structure group ensures that the two bundles are isomorphic.

Proposition 1.3.10 Every morphism of principal bundles is an isomorphism of principal bundles.

Proposition 1.3.11 Let G be topological group. A principal G -bundle is trivial if and only if it admits a global section.

This is entirely untrue for general bundles. For examples, the zero section of a fiber bundle is a global section.

1.4 The Diagonal Action

Definition 1.4.1 (The Diagonal Construction)

Let X be a space. Let G be a topological group. Let P be a principal G -bundle over X . Let F be a space with a continuous action of G . Define the associated fiber bundle to be the orbit space

$$P \times_G F = \frac{P \times F}{G}$$

where G acts on $P \times F$ by $g \cdot (e, f) = (g \cdot e, g \cdot f)$.

1.5 Fiber Bundle Construction Theorem

Let X be a space. Let G be a topological group. Recall that the fiber bundle construction theorem gives a correspondence of data

$$\{E \rightarrow B \mid E \text{ is a } G\text{-bundle with fiber } F\} \xleftrightarrow{1:1} \left\{ \begin{array}{l} \text{An open cover } \{U_i \mid i \in I\} \text{ of } B \\ \text{Maps } \{g_{ij} : U_i \cap U_j \rightarrow G \mid i, j \in I\} \text{ satisfying the cocycle condition} \\ \text{A space } F \text{ such that } G \text{ acts continuously on } F \end{array} \right\}$$

Notice that the data on the right comes in two parts. First an open cover $\{U_i \mid i \in I\}$ of B and maps $g_{ij} : U_i \cap U_j \rightarrow G$ satisfying the cocycle condition. Then a space F with a continuous action of G . We show that constructing the associated fiber bundle E_g is a two step process involving a universal object which is the principal G -bundle.

Definition 1.5.1 (Associated Principal Bundles)

Let X be a space. Let G be a topological group. Let $\{U_i \mid i \in I\}$ be an open cover of B . Let $\{g_{ij} : U_i \cap U_j \rightarrow G \mid i, j \in I\}$ be a collection of map satisfying the cocycle condition. Define the associated principal bundle to be the space

$$P(g_{ij}) = \frac{\coprod_{i \in I} U_i \times G}{\sim}$$

where $(x, h) \sim (x, g_{ij}(x)h)$ for all $x \in U_i \cap U_j$ and $h \in G$.

Definition 1.5.2 (Associated Fiber Bundle)

Let X be a space. Let G be a topological group. Let $\{U_i \mid i \in I\}$ be an open cover of B . Let $\{g_{ij} : U_i \cap U_j \rightarrow G \mid i, j \in I\}$ be a collection of map satisfying the cocycle condition. Let F be a space with a continuous action of G . Define the associated fiber bundle to be the space

$$E(g_{ij}, F) = \frac{\coprod_{i \in I} U_i \times F}{(x, f) \sim (x, g_{ij}(x)f)}$$

for $x \in U_i \cap U_j$ and $f \in F$.

Proposition 1.5.3

Let X be a space. Let G be a topological group. Let $\{U_i \mid i \in I\}$ be an open cover of B . Let $\{g_{ij} : U_i \cap U_j \rightarrow G \mid i, j \in I\}$ be a collection of map satisfying the cocycle condition. Let F be a

space with a continuous action of G . Then there is an isomorphism of fiber bundles

$$P(g_{ij}) \times_G F \cong E(g_{ij}, F)$$

Theorem 1.5.4 (Fiber Bundle Construction Theorem)

Let X be a space. Let G be a topological group. Then there is a one to one correspondence

$$\left\{ \begin{array}{l} \text{An open cover } \{U_i \mid i \in I\} \text{ of } X \\ \text{Maps } \{g_{ij} : U_i \cap U_j \rightarrow G \mid i, j \in I\} \\ \text{satisfying the cocycle condition} \end{array} \right\} \times \{F \in \mathbf{Top} \mid F \text{ has a continuous action of } G\} \xrightarrow{1:1} \{E \rightarrow X \mid E \text{ is a } G\text{-bundle with fiber } F\}$$

given by $(P(g_{ij}), F) \mapsto P \times_G F$.

Just as the structure groups and trivialization charts determine the isomorphism type of a fiber bundle, the $\widetilde{g}_{ij}$ and a map of base space $h : B_1 \rightarrow B_2$ completely determines a morphism of G -bundle.

Theorem 1.5.5

Let (F, E_1, B_1) and (F, E_2, B_2) be two G -bundles for a topological group G with the same fiber F . Suppose that we have the following.

- A map $h : B_1 \rightarrow B_2$ of base space
- A collection $\{\widetilde{g}_{ij} : U_i \cap h^{-1}(V_j) \rightarrow G \mid i \in I, j \in J\}$ of continuous maps such that

$$\begin{aligned} \widetilde{g}_{jk}(x) \cdot g_{ij}(x) &= \widetilde{g}_{ik}(x) & \text{for all } x \in U_i \cap U_j \cap h^{-1}(V_k) \\ g'_{jk}(h(x)) \cdot \widetilde{g}_{ij}(x) &= \widetilde{g}_{ik}(x) & \text{for all } x \in U_i \cap h^{-1}(V_j \cap V_k) \end{aligned}$$

Then there exists a unique G -bundle morphism having h as the map of base space and having $\{\widetilde{g}_{ij} \mid i, j \in I\}$ as its mapping transformations.

2 Bundle Structures in Equivariant Topology

2.1 Basic Properties

Proposition 2.1.1

Let G be a topological group. Let H be a closed subgroup of G . Then the quotient map $G \rightarrow G/H$ is a fiber bundle with fiber H and structure group $N_G(H)/H$.

2.2 The Bundle Structure Theorem

Theorem 2.2.1 (The Bundle Structure Theorem) Let B be a topological group and let G be a closed subgroup of B . Let H be a closed subgroup of G . Suppose that B/G has a local cross section at $G \in B/G$. Then the induced map of cosets

$$p : B/H \rightarrow B/G$$

give a G/H_0 -bundle with fiber G/H , where

$$H_0 = \bigcap_{g \in G} gHg^{-1}$$

Theorem 2.2.2 Let B be a topological group and let G be a closed subgroup of B . Let H be a closed subgroup of G . Then any two local cross sections of B/G at $G \in B/G$ induces equivalent bundles.

Corollary 2.2.3 Let B be a topological group and let G be a closed subgroup of B . Then the projection map $p : B \rightarrow B/G$ is a G -bundle with fiber G .

2.3 Reduction of the Structure Group

3 Classifying Spaces

3.1 Milnor's Join Construction

Definition 3.1.1 (The Join of two Spaces)

Let X, Y be spaces. Define the join of X and Y to be

$$X * Y = \frac{X \times I \times Y}{\sim}$$

where $(x, 0, y_1) \sim (x, 0, y_2)$ for $x \in X$ and $y_1, y_2 \in Y$ and $(x_1, 1, y) \sim (x_2, 1, y)$ for $x_1, x_2 \in X$ and $y \in Y$.

Example 3.1.2

Let X be a space. Then the following are true.

- $X * \{p\} = CX$.
- $X * \{p_1, p_2\} = \Sigma X$.

Example 3.1.3

Let $n, m \in \mathbb{N}$. Then we have

$$S^n * S^m \cong S^{n+m+1}$$

Example 3.1.4

We have $x_0 * \cdots * x_n \cong \Delta^n$.

Lemma 3.1.5

Let X, Y be spaces. Then the inclusions $i_X, i_Y : X, Y \rightarrow X * Y$ are null homotopic.

Definition 3.1.6 (Milnor's Join Construction)

Let G be a topological group. Define the Milnor join construction of G to be

$$EG = \lim_{k \rightarrow \infty} G^{*(k+1)}$$

Lemma 3.1.7

Let G be a topological group. Then the following are true.

- EG is an G -equivariant CW complex.
- EG is contractible.
- The projection map $EG \rightarrow EG/G$ is a universal principle G -bundle.

3.2 Universal Principal Bundles

Recall that homotopic maps give isomorphic fiber bundles. (??? Descends to isomorphic principal G -bundles)

Definition 3.2.1 (Principal Bundle Functor) Let G be a topological group and X a space. Define a contravariant functor $\text{Prin}_G : \mathbf{hTop} \rightarrow \mathbf{Set}$ as follows.

- For X a topological space, $\text{Prin}_G(X)$ is the set of isomorphism classes of principal G -bundles over X .
- If $[f : X \rightarrow Y]$ is a homotopy class of continuous maps,

$$\text{Prin}_G([f]) : \text{Prin}_G(Y) \rightarrow \text{Prin}_G(X)$$

is defined as follows. If $[p : E \rightarrow Y]$ is an homotopy class of principal G -bundles over Y , then it is sent to $[f^*(E)]$ the homotopy class of the pullback of p .

Proposition 3.2.2

Let G be a topological group. Then the principal bundle functor restricted to CW complexes is representable via the nerve of the category BG . Explicitly, this means that there exists a principal G -bundle $EG \rightarrow BG$ together with a natural isomorphism

$$\psi : [X, BG] \rightarrow \text{Prin}_G(X)$$

This natural isomorphism is defined by $f \mapsto [f^*(EG)]$. (Milnor's Join Construction)

Note that the theorem is still true when the principal bundle functor is restricted to paracompact spaces (with the same universal object).

Definition 3.2.3 (Universal G -Bundles)

Let G be a topological group. A principal G -bundle (F, E, B) is said to be universal if it represents the principal bundle functor.

Theorem 3.2.4

Let (F, E, B) be a principal G -bundle. If E is contractible then (F, E, B) is a universal G -bundle.

A surprising thing is that BG is not determined by its isomorphism type but instead by the weaker condition of its homotopy type.

Proposition 3.2.5

Let G be a topological group. Let $p : E \rightarrow B$ be a universal G -bundle. Then the following are true.

- B is weakly equivalent to a CW complex (hence WLOG we may always take the base space of a universal G -bundle to be a CW complex).
- Let $p : E \rightarrow B$ be a universal G -bundle such that B is a CW complex. Then B is unique up to canonical homotopy equivalence.
- E is unique up to G -homotopy equivalence.

3.3 Properties of the Classifying Space

Proposition 3.3.1

Let G be a topological group. Let BG be the classifying space of G . Then there is a homotopy equivalence

$$G \simeq \Omega BG$$

Corollary 3.3.2

Let G be a topological group. Then there is a natural bijection

$$\text{Prin}_G(S^n) \cong \pi_{n-1}(G)$$

Corollary 3.3.3

Let G be a discrete topological group. Then the classifying space BG is a $K(G, 1)$ -space.

Example 3.3.4

There is a homotopy equivalence $B\mathbb{Z} \simeq S^1$.

Example 3.3.5

There are homotopy equivalences

$$K(\mathbb{Z}/2\mathbb{Z}, 1) \simeq B\mathbb{Z}/2\mathbb{Z} \simeq \mathbb{RP}^\infty$$

Proposition 3.3.6

Let G, H be topological groups. Then there is a homotopy equivalence

$$B(G \times H) \simeq BG \times BH$$

Corollary 3.3.7

Let G, H be topological groups. Let X be a space. Then there is a natural bijection

$$\text{Prin}_{G \times H}(X) \cong \text{Prin}_G(X) \times \text{Prin}_H(X)$$

3.4 Relation to Vector Bundles**Definition 3.4.1 (Frame Bundle)**

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Define the frame bundle $F(E)$ of p to be the associated bundle of p .

Note that this is not the same as the associated vector bundle of p , which is isomorphic to p itself. Explicitly, the total space of the frame bundle of a vector bundle $p : E \rightarrow B$ of rank n is given by

$$F(E) = \frac{\coprod_{i \in I} U_i \times \text{GL}_n(F)}{\sim}$$

where $(i, b, v) \sim (j, b, g_{i,j}(b)(v))$ and $g_{i,j} : U_i \cap U_j \rightarrow \text{GL}_n(F)$ are the transition functions.

Since vector bundles are just fiber bundles with structure group the general linear group, it follows that frame bundles are principal bundles over their structure group.

Proposition 3.4.2

Let X be a space. Then the following are true.

- There is a natural bijection

$$\phi : \text{Prin}_{\text{GL}(n, \mathbb{R})}(X) \xrightarrow{\cong} \text{Vect}_n^{\mathbb{R}}(X)$$

given by mapping $p : E \rightarrow B$ to the frame bundle $F(E)$.

- There is a natural bijection

$$\phi : \text{Prin}_{\text{GL}(n, \mathbb{C})}(X) \xrightarrow{\cong} \text{Vect}_n^{\mathbb{C}}(X)$$

given by the same map.

Proposition 3.4.3

Let X be a paracompact space. Then the following are true.

- There is a natural bijection

$$\phi : \text{Prin}_{O(n)}(X) \xrightarrow{\cong} \text{Vect}_n^{\mathbb{R}}(X)$$

given by mapping $p : E \rightarrow B$ to the frame bundle $F(E)$.

- There is a natural bijection

$$\phi : \text{Prin}_{U(n)}(X) \xrightarrow{\cong} \text{Vect}_n^{\mathbb{C}}(X)$$

given by the same map.

Example 3.4.4

The following are true.

- The only non-trivial line bundle of $\mathbb{RP}^n$ is given by the universal line bundle γ_1 .
- The only non-trivial line bundles of $\mathbb{CP}^n$ is given by $\gamma_1^{\otimes n}$ or $(\gamma_1^*)^{\otimes n}$ for some $n \geq 0$ where γ_1

is the universal line bundle.

Proposition 3.4.5

Let $n \in \mathbb{N}$. Then following are true.

- There is a homotopy equivalence

$$BGL(n, \mathbb{R}) \simeq BO(n)$$

- There is a homotopy equivalence

$$BGL(n, \mathbb{C}) \simeq BU(n)$$

Proposition 3.4.6

Let $n \in \mathbb{N}$. Then the following are true.

- We have $BO(n) \cong \text{Gr}_n(\mathbb{R}^\infty)$.
- We have $BU(n) \cong \text{Gr}_n(\mathbb{C}^\infty)$.

Proposition 3.4.7 The following are true regarding the classifying space of the limits of the orthogonal and unitary groups.

- There is a homeomorphism $BO \cong \lim_{n \rightarrow \infty} BO(n)$
- There is a homeomorphism $BU \cong \lim_{n \rightarrow \infty} BU(n)$

Example 3.4.8

The following are true.

- The only non-trivial line bundle of $\mathbb{R}P^n$ is given by the universal line bundle γ_1 .
- The only non-trivial line bundles of $\mathbb{C}P^n$ is given by $\gamma_1^{\otimes n}$ or $(\gamma_1^*)^{\otimes n}$ for some $n \geq 0$ where γ_1 is the universal line bundle.

4 Characteristic Classes

4.1 Characteristic Classes as a Ring

Definition 4.1.1 (Characteristic Classes) Let G be a topological group and X a space. Denote $\text{Prin}_G(X)$ the isomorphism classes of principal G -bundles over X . Let $H^*(-)$ be a cohomology functor. A characteristic class for G is a natural transformation

$$c : \text{Prin}_G(-) \Rightarrow H^*(-)$$

Explicitly, if $p : E \rightarrow X$ is a principal G -bundle, then c assigns p to a cohomology class $c(p) \in H^*(X)$.

Example 4.1.2

Let G be a topological group. Suppose that c is a characteristic class. Let B be a space. Suppose that $\varepsilon : B \times G \rightarrow B$ is the trivial principal G -bundle. Then we have

$$c(\varepsilon) = 1$$

Definition 4.1.3 (Ring of Characteristic Classes) Let G be a topological group. Let R be a commutative ring. Define $\text{Char}_G(R)$ to be the set of all characteristic classes for principal G -bundles that take values in $H^*(-; R)$.

Proposition 4.1.4 Let G be a topological group. Let R be a commutative ring. Then $\text{Char}_G(R)$ is a ring with unit the constant characteristic class.

Theorem 4.1.5 Let G be a topological group and let R be a commutative ring. Then there is an isomorphism

$$\text{Char}_G(R) \cong H^*(BG; R)$$

4.2 The Stiefel-Whitney Class

Definition 4.2.1 (The Stiefel-Whitney Class) Consider the group $O(n)$. Let

$$w_i : \text{Prin}_{O(n)}(-) \rightarrow H^i(-, \mathbb{Z}/2\mathbb{Z})$$

be a collection of natural transformations. We say that they form a Stiefel-Whitney class if the following are satisfied.

1. Rank: If E is a principal $O(n)$ -bundle, then $w_0(E) = 1$ and $w_i(E) = 0$ for $i > \text{rank}(E)$.
2. Naturality: Let $p : E \rightarrow X$ be a principal $O(n)$ -bundle and let $f : Y \rightarrow X$ be a map. Then

$$w_i(f^*(E)) = f^*(w_i(E))$$

3. Whitney Product Formula: If E_1, E_2 are principal $O(n)$ -bundles, then

$$w_k(E_1 \oplus E_2) = \sum_{i=0}^k w_i(E_1) \smile w_{k-i}(E_2)$$

4. Normalization: If γ is the tautological line bundle over $\mathbb{P}^1(\mathbb{R})$, then $w_1(\gamma)$ is non-zero.

Proposition 4.2.2

The Stiefel-Whitney class exists and is unique.

Definition 4.2.3 (The Total Stiefel-Whitney Class)

Define the total Stiefel-Whitney class to be the natural transformation

$$w = \sum_{k=0}^{\infty} : \text{Prin}_{O(n)}(-) \Rightarrow H^*(-, \mathbb{Z}/2\mathbb{Z})$$

Lemma 4.2.4

Let E_1, E_2 be principal $O(n)$ bundles. Then we have

$$w(E_1 \oplus E_2) = w(E_1)w(E_2)$$

Proposition 4.2.5 The following are true regarding the Stiefel-Whitney class.

- If $p_1 : E_1 \rightarrow B_1$ and $p_2 : E_2 \rightarrow B_2$ are isomorphic principal $O(n)$ -bundles, then $w(E_1) = w(E_2)$
- If $e = B \otimes \mathbb{R}^n$ is the trivial bundle, then $w(e \oplus E) = w(E)$ for any principal $O(n)$ -bundle E .
-

Proposition 4.2.6

Let $n \in \mathbb{N}^+$. Then there is a natural isomorphism

$$\text{Char}_{O(n)}(\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[x_1, \dots, x_n]$$

where $\deg(x_i) = 1$ and the isomorphism is given by $x_i \mapsto w_i$ the i th Stiefel-Whitney class.

Proposition 4.2.7 Let X be a space. Then the function

$$w_1 : \text{Vect}_1^{\mathbb{R}}(X) \rightarrow H^1(X; \mathbb{Z}/2\mathbb{Z})$$

is a homomorphism. It is an isomorphism if X is homotopy equivalent to a CW complex.

Corollary 4.2.8 Let X be homotopy equivalent to a CW complex. Then any vector bundle $E \rightarrow X$ is orientable if and only if $w_1(E) = 0$.

Proposition 4.2.9

Let E be a principle $O(n)$ -bundle. If E has k nowhere vanishing sections, then we have $w_{n-k+1}(E) = \dots = w_n(E) = 0$

4.3 The Chern Class

Definition 4.3.1 (The Chern Class) Consider the group $U(n)$. Let

$$c_i : \text{Prin}_{U(n)}(-) \rightarrow H^{2i}(-, \mathbb{Z})$$

be a collection of natural transformations. We say that they form a Chern class if the following are satisfied.

1. Rank: If E is a principal $U(n)$ -bundle, then $c_0(E) = 1$ and $c_i(E) = 0$ for $i > \text{rank}(E)$.
2. Naturality: Let $p : E \rightarrow X$ be a principal $O(n)$ -bundle and let $f : Y \rightarrow X$ be a map. Then

$$c_i(f^*(E)) = f^*(c_i(E))$$

3. Whitney Product Formula: If E_1, E_2 are principal $O(n)$ -bundles, then

$$c_k(E_1 \oplus E_2) = \sum_{i=0}^k c_i(E_1) \smile c_{k-i}(E_2)$$

4. Normalization: If γ is the tautological line bundle over $\mathbb{P}^1(\mathbb{R})$, then $c_1(\gamma)$ is non-zero.

Proposition 4.3.2 The Chern class exists and is unique.

Definition 4.3.3 (The Total Chern Class)

Define the total Chern class to be the natural transformation

$$c = \sum_{k=0}^{\infty} c_k : \text{Prin}_{O(n)}(-) \rightarrow H^*(-, \mathbb{Z})$$

Lemma 4.3.4

Let E_1, E_2 be principal $U(n)$ bundles. Then we have

$$c(E_1 \oplus E_2) = c(E_1)c(E_2)$$

Proposition 4.3.5

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{C}$. Then $c_1(E) = 0$ if and only if E admits a reduction of structure group to $SU(n)$.

TBA: First chern class is complete invariant of complex line bundles. First Stiefel-Whitney class is a complete invariant of real line bundle.

Proposition 4.3.6 Let $n \in \mathbb{N}^+$. Then there is a natural isomorphism

$$\text{Char}_{U(n)}(\mathbb{Z}) \cong \mathbb{Z}[x_1, \dots, x_n]$$

where $\deg(x_i) = 2$ and the isomorphism is given by $x_i \mapsto c_i$ the i th Chern class.

Combined with theorem 5.1.3, this means that we can now compute the cohomology of the classifying space of $U(n)$. Namely, we have an isomorphism

$$H^*(BU(n); \mathbb{Z}) \cong \mathbb{Z}[c_1, \dots, c_n]$$

Proposition 4.3.7

Let X be a space. If X is homotopy equivalent to a CW complex, then there is a natural isomorphism

$$c_1 : \text{Vect}_1^{\mathbb{C}}(X) \xrightarrow{\cong} H^2(X; \mathbb{Z})$$

Proof

Since $\mathbb{CP}^{\infty}$ is a $K(\mathbb{Z}, 2)$ -space, we deduce that

$$\text{Vect}_1^{\mathbb{C}}(X) \cong \text{Prin}_{U(1)}(X) \cong [X, BU(1)] \cong [X, \mathbb{CP}^{\infty}] \cong H^2(X; \mathbb{Z})$$

and one can check that this bijection respects group structure. ■

Theorem 4.3.8 (The Splitting Principle)

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{C}$. Then there exists a space $F(B)$ and a map $\pi : F(B) \rightarrow B$ such that the following are true.

- $\pi^* : H^*(B; \mathbb{Z}) \rightarrow H^*(F(B); \mathbb{Z})$ is injective.
- $\pi^*(E)$ is a direct sum of line bundles.

4.4 The Pontryagin Class

Definition 4.4.1 (The Pontryagin Class)

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over $\mathbb{R}$. Define the Pontryagin class of E to be

$$p_i(E) = (-1)^i c_{2i}(E \otimes_{\mathbb{R}} \mathbb{C}) \in H^{4i}(-, \mathbb{Z})$$

Definition 4.4.2 (The Total Pontryagin Class)

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over $\mathbb{R}$. Define the total Pontryagin class of E to be

$$p(E) = \sum_{i=0}^{\infty} p_i(E)$$

Proposition 4.4.3

Let B be a space. Let E_1 and E_2 be two vector bundles over $\mathbb{R}$. Then we have

$$2(p(E_1 \oplus E_2) - p(E_1)p(E_2)) = 0 \in H^*(B, \mathbb{Z})$$

5 More on Orientations of Vector Bundles

5.1 The Orientation Cover

Lemma 5.1.1

Let V be a finite dimensional vector space over $\mathbb{R}$. Then there is a bijection

$$\{[w] \mid [w] \text{ is an orientation of } V\} \xrightarrow{1:1} H^n(V, V \setminus \{0\}; \mathbb{Z})$$

Definition 5.1.2 (The Orientation Cover)

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Define the orientation cover to be the space

$$\tilde{E} = \{(b, w_b) \mid b \in B, w_b \text{ is an orientation of } E_b\}$$

Since orientations are in bijection with $H^n(V, V \setminus \{0\}; \mathbb{Z})$, we may equivalently define the orientation cover using elements of the top cohomology of the fiber bundle. This viewpoint is useful in generalizing orientation to arbitrary coefficient rings.

Lemma 5.1.3

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Then $\tilde{E}$ is a covering space of B .

Proposition 5.1.4

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Then the following are equivalent.

- E is orientable.
- E admits a reduction of structure groups to $\mathrm{SL}(n, \mathbb{R})$.
- $\mathrm{Fr}(E) \times_{\mathrm{GL}(n, \mathbb{R})} \{\pm 1\}$ admits a global section.
- $w_1(E) = 0$.
- $\Lambda^n(E)$ is isomorphic to the trivial bundle.
- $\tilde{E}$ admits a global section.
- $\tilde{E}$ is isomorphic to the trivial bundle.
- There exists a class $u \in H^n(E, E \setminus B; \mathbb{Z})$ where B is identified with the zero section of E , such that the image of u in the map $H^n(E, E \setminus B; \mathbb{Z}) \rightarrow H^n(E_b, E_b \setminus \{0\}; \mathbb{Z}) \cong \mathbb{Z}$ induced by inclusion is a generator.

Definition 5.1.5 (The R -Orientation Cover)

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Let R be a ring. Define the R -orientation cover of E to be

$$\tilde{E}_R = \{(b, w_b) \mid b \in B, w_b \in H^n(E_b, E_b \setminus \{0\}; R) \cong R \text{ is a generator}\}$$

Lemma 5.1.6

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Let R be a ring. Then $\tilde{E}_R$ is a covering space of B .

Definition 5.1.7 (R -Orientability)

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Let R be a ring. We say that E is R -orientable if E admits a global section.

Definition 5.1.8 (The Thom Class)

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Let R be a ring. Let

$s_0 : b \rightarrow E$ denote the zero section. Define the Thom class of E to be any class

$$u \in H^n(E, E \setminus s_0(B); R)$$

such that the image of u in the map $H^n(E, E \setminus s_0(B); R) \rightarrow H^n(E_b, E_b \setminus \{0\}; R) \cong R$ induced by the inclusion is a generator.

Lemma 5.1.9

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Let R be a ring. Then the following are equivalent.

- E is R -orientable.
- $\tilde{E}$ is isomorphic to the trivial bundle.
- E admits a Thom class.

Corollary 5.1.10

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{R}$. Then E is $\mathbb{Z}/2\mathbb{Z}$ -orientable.

5.2 The Euler Class

Definition 5.2.1 (The Euler Class)

Let B be a space. Let $p : E \rightarrow B$ be an orientable vector bundle of rank n with fiber $\mathbb{R}$. Let $s_0 : B \rightarrow E$ be the zero section. Let $u \in H^n(E, E \setminus s_0(B); \mathbb{Z})$ be a Thom class of E . Define the Euler class of E to be

$$e(E) = s_0^*(u) \in H^n(B; \mathbb{Z})$$

Proposition 5.2.2

Let B be a space. Let $p : E \rightarrow B$ be an orientable vector bundle with fiber $\mathbb{R}$. Then the following are true.

- If $f : X \rightarrow B$ is a map, then $e(f^*(E)) = f^*(e(E))$.
- If $q : F \rightarrow B$ is an orientable vector bundle, then $e(E \oplus F) = e(E) \cup e(F)$.

Corollary 5.2.3

Let B be a space. Let $p : E \rightarrow B$ be an orientable vector bundle of rank n with fiber $\mathbb{R}$. If n is odd, then $e(E) \cup e(E) = 0$.

Proposition 5.2.4

Let B be a space. Let $p : E \rightarrow B$ be an orientable vector bundle of rank n with fiber $\mathbb{R}$. Let $f : H^n(B; \mathbb{Z}) \rightarrow H^n(B; \mathbb{Z}/2\mathbb{Z})$ be the change of rings homomorphism. Then we have $f(e(E)) = w_n(E)$.

Proposition 5.2.5

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle of rank n with fiber $\mathbb{R}$. If E admits a nowhere vanishing section, then $e(E) = 0$.

5.3 Classification of Characteristic Classes of Oriented Vector Bundles

def of tautological bundle γ_{2n} over $BSO(2n)$.

Proposition 5.3.1

Let R be an integral domain such that 2 is invertible. Let $n \in \mathbb{N}$. Then we have

$$H^*(BSO(2n+1); R) \cong R[p_1, \dots, p_n] \quad \text{and} \quad H^*(BSO(2n); R) \cong R[p_1, \dots, p_n, e(\gamma_{2n})]$$

6 Thom Space and the Thom Isomorphism Theorem

6.1 Thom Space

Definition 6.1.1 (Unit Sphere and Unit Disc Bundle) Let $p : E \rightarrow B$ be an n -dimensional vector bundle over $\mathbb{R}$. Let $\langle -, - \rangle : E \times E \rightarrow \mathbb{R}$ be a smoothly varying inner product on E . Define the disc bundle to be

$$D(E) = \{e \in E \mid \langle e, e \rangle \leq 1\}$$

together with the map $p|_{D(E)} : D(E) \rightarrow B$. Define the sphere bundle to be

$$S(E) = \{e \in E \mid \langle e, e \rangle = 1\}$$

together with the map $p|_{S(E)} : S(E) \rightarrow B$.

Definition 6.1.2 (Thom Space) Let $p : E \rightarrow B$ be an n -dimensional vector bundle over $\mathbb{R}$ such that B is paracompact. Define the Thom space of E to be

$$T(E) = \frac{D(E)}{S(E)}$$

The base point is taken as the equivalent class $S(E)$ if needed.

Lemma 6.1.3 Let $p : E \rightarrow B$ be an n -dimensional vector bundle over $\mathbb{R}$. Let E_0 denote the zero section of E . Then there is a natural isomorphism

$$\tilde{H}^n(T(E); G) = H^n(E, E \setminus E_0)$$

for any abelian group G .

Theorem 6.1.4 (The Thom Isomorphism) Let $p : E \rightarrow B$ be an n -dimensional vector bundle over $\mathbb{R}$. Let E_0 denote the zero section of E . Then there exists a unique $u \in H^n(E, E \setminus E_0; \mathbb{Z}/2\mathbb{Z})$ such that

$$u|_{(F_b, F_b \setminus \{0\})} \in H^n(F_b, F_b \setminus \{0\}; \mathbb{Z}/2\mathbb{Z})$$

is non-zero for all $b \in B$. Moreover, there is an isomorphism

$$\Phi : H^k(E; \mathbb{Z}/2\mathbb{Z}) \rightarrow \tilde{H}^{k+n}(E, E \setminus E_0; \mathbb{Z}/2\mathbb{Z}) \cong \widetilde{H}^n(T(E); \mathbb{Z}/2\mathbb{Z})$$

given by $y \mapsto y \smile u$ for all $k \in \mathbb{Z}$.

Ref: Milnor

7 Cohomology of Fiber Bundles

Theorem 7.0.1 (Leray-Hirsch Theorem) Let $p : E \rightarrow B$ be a fiber bundle with fiber F . Suppose the for some ring R , the following are true.

- $H^n(F; R)$ is a finitely generated free R -module for all $n \in \mathbb{N}$
- For each fiber F and inclusion $i : F \rightarrow E$, there exists $c_j \in H^{k_j}(E; R)$ such that the collection $i^*(c_j) \in H^{k_j}(F; R)$ form a basis for the free R -module $H^*(F; R)$

Then the map

$$\Phi : H^*(B; R) \otimes_R H^*(F; R) \rightarrow H^*(E; R)$$

defined by $\sum_{i,j} b_i \otimes_R i^*(c_j) \mapsto \sum_{i,j} p^*(b_i) \smile c_j$ is an isomorphism.